- *Low-observable and self-healing materials*: Nano-composite skins and RAM coatings with embedded microcapsules or fiber meshes that support limited in-flight self-repair and adaptive surface morphing for drag/stealth optimization.
- *Redundant avionics and FDIR sensors*: Multi-lane flight control units, redundant IMUs/GNSS receivers, independent power rails, and health sensors for vibration, strain, and thermal state to enable prognostics and predictive maintenance.
- *Modular payload bays and interfaces*: Standardized mechanical/electrical payload interfaces (power, data, mechanical latches) that allow swap-in of mission modules (weapons, comms relays, scientific instruments) with minimal integration time.
- *Physical safety and recovery systems*: Parachutes, precision auto-landing aids (vision-based or RTK anchors), or net/capture interfaces designed to reduce risk to people and infrastructure during failure events.

6.3.3 Major software and systems components

The software architecture is multi-layered, modular, and certifiable, where safety and accountability are required. Core components include:

- *Autonomy/mission manager*: Hierarchical planners strategic (policy, mission objectives), operational (task allocation, coalition planning), and tactical (trajectory generation, engagement decisions). Supports plan re-synthesis under new constraints.
- *Perception and sensor fusion stack*: Time-synchronized pipelines combining radar, EO/IR, LiDAR, SIGINT, and navigation sensors into fused tracks and semantic maps. Uses probabilistic filters (e.g.,